

LOON LAKE, SK, Canada

WMO: 712730

Lat: 54.0197N

Lon: 109.1383W

Elev: 1790

StdP: 13.77

Time Zone: -6.00 (W06)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.3	-22.5	-34.9	0.8	-27.8	-28.9	1.2	-22.1	16.5	17.2	14.4	14.2	2.4	280	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	80.9	62.6	78.0	61.6	75.4	60.4	65.8	76.6	64.0	74.4	62.3	71.8	7.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.6	87.6	70.8	59.9	82.3	68.9	58.3	77.6	66.9	31.6	76.5	30.2	74.6	28.9	71.7	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.2	12.6	DB	-36.1	86.0	5.4	2.0	-39.9	87.5	-43.1	88.7	-46.1	89.8	-50.0	91.3	(4)
(5)			WB	-36.2	70.4	5.4	2.5	-40.1	72.2	-43.3	73.6	-46.3	75.0	-50.2	76.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	8.8	8.0	21.9	36.5	50.8	58.4	62.6	59.9	51.5	38.0	21.3	8.4		
	DBStd	23.10	16.33	15.08	14.54	9.86	8.23	5.59	4.42	5.19	8.07	8.66	12.63	15.32		
	HDD50	6451	1276	1177	870	414	93	7	0	3	79	381	862	1289		
	HDD65	10761	1741	1597	1335	856	444	207	101	168	408	838	1312	1754		
	CDD50	1215	0	0	0	9	117	258	390	310	123	8	0	0		
	CDD65	50	0	0	0	0	3	8	26	11	2	0	0	0		
	CDH74	871	0	0	0	3	115	136	312	202	103	0	0	0		
	CDH80	119	0	0	0	0	15	16	45	21	22	0	0	0		
	WSAvg	5.7	5.5	5.6	6.2	6.8	6.3	5.7	5.2	4.7	5.4	5.9	5.8	5.3		
	PrecAvg	17.4	0.6	0.5	0.8	0.9	1.7	3.3	2.9	2.6	1.9	0.6	0.6	0.7		
(15) Precipitation	PrecMax	23.6	1.7	1.4	2.4	2.6	6.4	9.6	7.4	6.4	4.4	1.6	1.9	2.0		
	PrecMin	12.6	0.1	0.0	0.1	0.0	0.1	1.0	0.7	0.2	0.4	0.0	0.0	0.2		
	PrecStd	3.0	0.4	0.4	0.6	0.6	1.3	1.9	1.4	1.4	1.0	0.4	0.5	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	44.4	55.6	72.3	82.0	82.3	84.7	82.7	83.6	68.2	52.9	43.0		
		MCWB	37.7	36.4	42.4	50.2	56.9	61.5	67.4	64.9	62.7	52.2	43.4	36.2		
	2%	DB	40.8	38.9	49.3	63.9	77.6	77.9	80.5	78.7	76.5	60.6	46.2	38.7		
		MCWB	35.2	33.3	39.1	45.9	55.0	60.5	65.5	64.1	59.5	48.1	38.7	33.3		
	5%	DB	37.5	34.7	44.2	58.5	72.5	74.5	77.4	76.0	71.1	55.9	41.0	34.4		
		MCWB	32.9	30.4	36.3	43.7	53.3	58.8	62.9	62.2	57.0	45.2	35.9	30.0		
	10%	DB	33.3	30.5	40.2	52.8	68.6	71.5	74.7	73.1	66.4	51.6	37.2	29.7		
		MCWB	29.8	27.9	34.1	40.9	51.5	57.0	61.7	60.5	54.8	43.9	33.4	26.9		
	0.4%	WB	38.1	37.7	43.1	50.4	59.7	65.0	70.0	68.1	63.6	53.7	44.3	36.9		
		MCDB	44.5	42.8	54.1	71.2	72.0	76.4	81.9	79.0	79.1	65.3	51.4	41.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.6	33.4	39.7	46.8	57.1	62.7	66.3	65.5	60.8	49.6	39.3	33.5		
		MCDB	40.6	38.4	48.0	61.9	71.6	73.5	77.3	75.7	74.7	56.3	45.1	38.6		
	5%	WB	33.2	30.9	36.8	44.4	54.8	60.8	64.7	63.5	58.1	46.7	36.1	30.1		
		MCDB	37.3	34.5	43.2	56.9	70.2	71.5	75.3	73.2	68.8	54.7	40.2	34.0		
	10%	WB	29.8	27.7	34.6	41.8	52.8	59.1	63.0	61.7	55.6	44.1	33.5	26.9		
		MCDB	33.4	30.8	39.7	51.6	66.5	68.4	72.0	70.8	65.0	51.3	36.7	29.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	20.5	19.5	19.7	24.8	20.7	20.8	21.5	21.6	16.8	14.5	16.4		
		MCDBR	19.2	21.3	23.1	30.0	32.6	26.7	26.2	27.4	31.3	25.4	19.8	19.6		
		MCWBR	15.0	16.3	14.6	16.0	14.6	12.4	13.1	14.6	16.0	14.8	12.8	15.3		
		MCDBR	19.3	20.5	20.7	28.1	29.0	22.1	24.4	25.0	29.8	23.9	17.8	18.9		
	5% WB	MCWBR	15.3	16.2	13.3	15.5	13.6	11.1	13.2	14.2	15.9	14.5	12.5	15.2		
(40) Clear-Sky Solar Irradiance	taub		0.238	0.258	0.294	0.320	0.344	0.363	0.365	0.361	0.309	0.281	0.249	0.213		
	taud		2.332	2.306	2.334	2.421	2.398	2.361	2.382	2.373	2.519	2.568	2.466	2.356		
	Ebn at Noon		240	271	283	290	286	281	278	272	277	264	238	232		
	Edh at Noon		17	24	30	32	35	36	35	33	25	19	15	13		
(44) All-Sky Solar Radiation	RadAvg		284	618	1101	1484	1810	1823	1872	1527	1070	611	275	201		
	RadStd		28	42	76	123	157	133	121	122	122	102	38	20		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page